COURSE TITLE : PETROLEUM & ENERGY ENGINEERING
COURSE CODE : 4074
COURSE CATEGORY : A
PERIODS/ WEEK : 4
PERIODS/ SEMESTER : 56
CREDIT : 4

TIME SCHEDULE

MODULE	TOPIC	PERIODS
1	Petroleum & nuclear fuels	14
2	Solid fuels and gaseous fuels	14
3	Analysis of fuel, Fuel burning system& furnaces	14
4	Non-conventional energy	14
TOTAL		**56**

COURSE OUTCOMES :

SL.NO	SUB	STUDENT WILL BE ABLE TO
1	1	know the important primary & secondary liquid fuels
	2	Understand the exploration, drilling and purification processes used in petroleum industry.
	3	Understand the nuclear fuels used for power generation.
2	4	Understand the characteristics of solid fuels
	5	Understand the coal beneficiation and coal carbonization processes.
	6	Understand the construction and working of calorimeters
	7	Analyze the energy content of various gaseous fuels.
3	8	Understand the analysis of coal & flue gases.
	9	Understand the construction and working of furnaces.
	10	Understand the burning system of coal& liquid fuels
4	11	Understand the energy production methods from non - conventional sources.

SPECIFIC OUTCOMES

MODULE – I PETROLEUM & NUCLEAR FUELS

1.1.0 Understand the Important primary and Secondary Liquids Fuels.

1.1.1 Define Petroleum, Origin, Formation of Petroleum, Exploration, and Drilling.
1.1.2 Name the different Crude Oil and their sources.
1.1.3 Classify Petroleum deposits
1.1.4 Explain fractional distillation of crude oil refining process with flow diagram

1.1.5 List the petroleum products from crude oil refining and their boiling ranges
1.1.6 Draw the fractionating column and name the parts
1.1.7 Explain the production of Gasoline by secondary process
1.1.8 Define cracking, list the types of cracking .
1.1.9 Explain with a flow diagram of fluidised catalytic cracking process
1.1.10 Explain with a flow diagram of fixed bed catalytic cracking process
1.1.11 Explain with a flow diagram of thermal cracking process .
1.1.12 Define hydro cracking, polymerisation, reforming, alkylation, isomerisation
1.1.13 Explain with a flow diagram of catalytic reforming
1.1.14 Explain with the flow diagram of catalytic polymerization
1.1.15 Explain with the flow diagram the Sulphuric acid alkylation process
1.1.16 List the types of isomerisation processes
1.1.17 Explain with the flow diagram the low temperature isomerisation process
1.1.18 Define octane number, knocking and cetaine number
1.1.19 Name the anti-knocking agents
1.1.20 Explain the various types of non petroleum liquid fuels
1.1.21 Describe the storage of liquid fuels

1.2.0 Understand the purification of petroleum products

1.2.1 Define sweetening of petroleum
1.2.2 List the important sweetening processes
1.2.3 Explain Doctor's sweetening process
1.2.4 Describe Copper Chloride sweetening process
1.2.5 Explain with flow diagram of Copper Chloride sweetening process for the sweetening of butane, gasoline, fuel oil and Kerosene

1.3.0 Understand the nuclear fuels used for power generation

1.3.1 Define Nuclear fusion reactors
1.3.2 Classify the Nuclear reactors for power generation
1.3.3 Draw the nuclear reactor & name the parts
1.3.4 List the nuclear fuel materials
1.3.5 Describe the occurrence of Uranium , Thorium& Plutonium
1.3.6 Explain magneto hydrodynamics power generation with the help of diagram
1.3.7 List the advantage of using hydrogen as an energy carrier
1.3.8 Describe the storage of hydrogen

1.4.0 Analysis of liquid fuel

1.4.1 Define flash and fire point of liquid fuel
1.4.2 Illustrate the function of Redwood viscometer
1.4.3 Illustrate the function of Pensky Marten's flash point apparatus

MODULE- II SOLID FUELS AND GASEOUS FUELS

2.1.0 Understand the characteristics of solid fuels

2.1.1 Define "fuel"
2.1.2 List primary & secondary solid fuels with examples
2.1.3 Explain the composition of wood
2.1.4 Explain the composition of charcoal
2.1.5 Explain the origin coal
2.1.6 List the theories behind origin of coal

2.1.7 Describe the ranks of coal
2.1.8 Describe the characteristics of peat
2.1.9 Describe the characteristics of Lignite
2.1.10 Describe the characteristics of semi bituminous
2.1.11 Describe the characteristics of sub bituminous
2.1.12 Describe the characteristics of Anthracite
2.1.13 Define fusion point of ash
2.1.14 Define calorific intensity
2.1.15 Describe the classification of Indian coals
2.1.16 Describe the proportion of coal for industrial purpose
2.1.17 Describe the properties of coal used for metallurgical process
2.1.18 Describe the properties of coal used for gas producers
2.1.19 Describe the properties of coal used for locomotive engines & steam generation

2.2.0 Understand the cleaning methods of coal

2.2.1 Explain gravity separation by wet & dry process
2.2.2 Explain froth floatation by wet process
2.2.3 Explain Jigbaum washing by wet process
2.2.4 Explain Wilfly table method by dry process
2.2.5 Explain Cyclone washing
2.2.6 Describe Storage of solid fuels
2.2.7 Define spontaneous combustion
2.2.8 Describe the precautions taken for storage of coal

2.3.0 Understand the different coal carbonization methods

2.3. 1 Name the two types of carbonization
2.3.2 Differentiate between low temperature & high temperature carbonization
2.3.3 Explain the manufacture coke using beehive oven
2.3.4 Explain the manufacture coke using Otto Hoffman by product oven.
2.3.5 State the properties of metallurgical coke.
2.3.6 Describe recovery of byproducts from coke manufacture
2.3.7 Describe the composition of coke oven gas
2.3.8 List the uses and calorific value of coke oven gas.

2.4.0 Understand the construction of various types of calorimeter

2.4.1 Define calorific value of fuel
2.4.2 Define higher & lower calorific value
2.4.3 List the apparatus used for determining calorific value
2.4.4 Draw the figure of Bomb calorimeter
2.4.5 Draw the figure of Junker's gas calorimeter
2.4.6 Explain construction, working & calculation of Bomb calorimeter
2.4.7 Explain construction,working & calculation of Junker's gas calorimeter
2.4.8 Define flame temperature
2..4.9 Define ignition temperature

2.5.0 Analyze the energy content of gaseous fuels

2.5.1 Define natural gas
2.5.2 State the calorific value, composition uses of natural gas.
2.5.3 Illustrate the manufacture of producer gas
2.5.4 List the uses of producer gas
2.5.5 State the reactions in the production process of producer gas
2.5.6 List the composition and calorific value of producer gas.

2.5.7 Explain the manufacture of water gas
2.5.8 State the reactions in the manufacture of water gas.
2.5.9 List the uses of water gas.
2.5.10 List the composition and calorific value of water gas.
2.5.11 Explain the manufacture of carbureted water gas with the aid of gas generator
2.5.12 List the uses of carbureted water gas.
2.5.13 List the composition & calorific value of carbureted water gas
2.5.14 Explain the manufacture of oil gas
2.5.15 List the composition and uses of oil gas
2.5.16 Describe the manufacture of coal gas with the help of diagram.
2.5.17 List the uses of coal gas
2.5.18 List the composition and calorific value of coal gas.
2.5.19 List the uses of LPG
2.5.20 State the composition & calorific value of LPG
2.5.21 Explain manufacture of Biogas/gobar gas
2.5.22 Draw the figure of biogas generator, name the parts
2.5.23 State the composition & calorific value of gobar gas
2..5.24 List the uses of biogas & gobar gas
2.5.25 List the composition of Blast furnace gas
2.5.26 List the calorific value of Blast furnace

MODULE III ANALYSIS OF FUEL, FUEL BURNING SYSTEMS AND FURNACES

3.1.0 Understand the analysis of coal & flue gases
3.1.1 Describe flue gas analysis
3.1.2 Name the apparatus used for flue gas analysis
3.1.3 Draw the figure & name the parts of Orsat apparatus
3.1.4 Explain construction,& working of Orsat apparatus
3.1.5 List the specific application of Orsat apparatus in industries
3.1.6 Name the analysis of coal
3.1.7 Define proximate & ultimate analysis
3.1.8 Describe the proximate analysis
3.1.9 Describe the ultimate analysis

3.2.0 Understand the construction & working of furnace
3.2.1 Define the term furnace
3.2.2 Explain the construction & working of blast furnace with the help of sketch
3.2.3 Explain the constructional details & working of open-hearth furnace with the help of a sketch.
3.2.4 Explain the construction and working of electric furnace.
3.2.5 Explain the different heat saving methods.
3.2.6 Explain regeneration waste heat recovery.

3.3.0 Understand the coal burning system.
3.3.1 List the different coal burning system.
3.3.2 Explain primary air amd secondary air supplied to the burning system.
3.3.3 Draw the figure and explain the working of over feed stocker.
3.3.4 Draw the figure and explain the working of traveling grate stocker.
3.3.5 Draw the figure and explain the working of chain grate stocker

3.3.6 Explain the coal burning system.
3.3.7 Explain the fluidized bed combustion system

3.4.0 Understand the liquid fuel burning system

3.4.1 Explain burning system for liquid fuel.
3.4.2 Draw the figure of the following oil atomizing burner, venturiburner, blast buner(low pressure and high pressure),rotary cap burner, swerling burner.
3.4.3 Explain burning system for gaseous fuel.
3.4.4 Draw the figure and explain inside mixing type burners.
3.4.5 Draw the figure and explain outside mixing type burners

MODULE- IV NON CONVENTIONAL ENERGY

4.1.0 Understand the methods of energy production from non-conventional sources

4.1.1 classify solar energy utilization
4.1.2 List the different types of collectors
4.1.3 Explain flat collectors with the help of diagram.
4.1.4 Explain liquid flat collectors with the help of sketch
4.1.5 Draw the cylindrical parabolic concentrating collectors.
4.1.6 Explain the working of cylindrical parabolic concentrating collectors
4.1.7 Draw the figure and explain the working solar heater
4.1.8 Describe the thermal applications of solar energy.
4.1.9 Draw the figure and explain water heating.
4.1.10 Draw the diagram of forced circulation water heating system.
4.1.11 Draw the diagram and name the parts of space heating.
4.1.12 Draw the schematic diagram solar air heater
4.1.13 Describe the space cooling and refrigeration
4.1.14 Draw the figure and name the parts of solar absorption refrigeration system
4.1.15 Explain the working solar still for using solar distillation with the help of figure
4.1.16 Evaluate the solar drying & solar cooking
4.1.17 Draw the diagram & name the parts of box type solar cooker
4.1.18 Describe the working of silicon cell
4.1.19 Evaluate the application of silicon cell
4.1.20 Draw the schematic diagram of photo voltaic water pumping system
4.1.21 Draw the diagram of satellite solar power station
4.1.22 Explain the working of wind mills
4.1.23 List the types of wind mill rotors
4.1.24 Draw the figure of different wind mill rotors
4.1.25 Evaluate the energy from Biomass
4.1.26 Describe principle working ocean thermal energy conversion
4.1.27 Draw the schematic diagram of OTEC plant
4.1.28 Explain the different sites of tidal energy in India
4.1.29 List the geothermal power stations in India
4.1.30 Explain tidal electric power generations

CONTENT DETAILS

MODULE – I PETROLEUM& NUCLEAR FUELS

Primary- secondary fuels-colloidal fuel-coal tar distillation-crude oil sources-classification of petroleum-fractional distillation-petroleum products-cracking- catalytic-,thermal-,hydro cracking-reforming-polymerisation-knocking agent-octane number-cetane number-removal of sulphur form gasoline,kerosene,LPG- storage of liquid fuels.

Nuclear fuels for power generation- fission reactors- the nuclear reactor and its components- fuel materials- fuel cycle- Classify the Nuclear reactors for power generation

MODULE – II SOLID FUELS AND GASEOUS FUELS

Solid fuels-Classification-Composition of wood-Charcoal-Origin of coal-theories of origin of coal-ranking of coal-characteristics-peat-lignite-sub bituminous coal-bituminous coal-semi anthracite coal-anthracite coal -classification of coal according to their analysis properties anduses of coal for industrial purpose-cleaning of coal-carbonization of coal –high temperature and low temperature carbonization-manufacture of coke-Beehive oven-OttoHoffman by product oven-storage of coal.

Calorific value-higher and lower calorific value-bomb calorimeter-Junker's gas calorimeter-flame temperature-Ignition temperature.

Gaseous fuel-Natural gas,composition and calorific value-manufacture,calorific value,Composition and uses of produer gas-water gas-carburreted water gas-oil gas-coal gas-bio gas/gober gas,uses,composition and calorific value of LPG and blast furnace gas.

MODULE – III ANALYSIS OF FUEL, FUEL BURNING SYSTEMS AND FURNACES

Analysis of coal-proximate and ultimate analysis-classification of coal according to their analysis, properties and uses of coal for industrial purpose-Classification of furnaces- construction and working of blast furnaces, open hearth, rotary kiln, electric arc furnaces – waste heat recovery-application of various furnaces-fuel burning system-stockers-grates- pulverized coal burning system-fluidized bed combustion system-burning system for liquid fuels- steam atomizing burners-burning system for gaseous fuel- inside and outside mixing burners

MODULE – IV NON CONVENTIONAL ENERGY

Solar energy-types of collectors – thermal application for solar energy-water heating – space heating- cooling and refrigeration- distillation-solar drying and cooking-working of silicon cell-Wind energy-working of wind mills energy from bio mass-working principle of ocean thermal energy conversion –Schematic diagram of OTEC-Geo thermal power- geo thermal power station in India-tidal energy sites in India-tidel energy power generation

REFERENCE

S.S.DARA	-	Text book of engineering Chemistry
Jain&Jain	-	Engineering chemistry
O.P.GUPTA	-	Fuel, furnaces and refractories
S.P.SUKHATME	-	Solar energy
IIT CHENNAI	-	Chem Tech Vol-I